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Stat 50

HW 7#

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Section 2.4

②

x	0	1	2	3	4
P(x)	0.4	0.3	0.15	0.10	0.05

(a) $P(x \leq 2)$:-

$$P(x \leq 2) = P(x=0) + P(x=1) + P(x=2)$$
$$= 0.4 + 0.3 + 0.15 = 0.85$$

$\boxed{P(x \leq 2) = 0.85}$

(b) $P(x > 1)$:

$$P(x > 1) = P(x=2) + P(x=3) + P(x=4)$$
$$= 0.15 + 0.10 + 0.05 = 0.30$$

$\boxed{P(x > 1) = 0.30}$

(c) μ_x :-

$$\mu_x = E(x) = \sum_x x\,P(x)$$
$$\mu_x = 0(0.4) + 1(0.3) + 2(0.15) + 3(0.10) + 4(0.05)$$
$$\mu_x = 0 + 0.30 + 0.30 + 0.30 + 0.2 = 1.1$$

$\boxed{\mu_x = 1.1}$

(d) σ_x^2

$$\sigma_x^2 = E(x^2) - (E(x))^2$$
$$E(x^2) = 0(0.4) + 1^2(0.3) + 2^2(0.15) + 3^2(0.10) + 4^2(0.05)$$
$$= 0 + 0.3 + 0.6 + 0.9 + 0.8 = 2.6$$
$$\sigma_x^2 = 2.6 - (1.1)^2 = 1.39$$

(2)

$\sigma_x^2 = 1.39$

(3)

(a)

x	1	2	3	4	5
P(x)	0.4	0.2	0.2	0.1	0.1

(i) $\mu_x = 1(0.4) + 2(0.2) + 3(0.2) + 4(0.1) + 5(0.1)$

$\mu_x = 0.4 + 0.4 + 0.6 + 0.4 + 0.5 = 2.3$

$\mu_x = 2.3$

(b) $E(x^2) = 1^2(0.4) + 2^2(0.2) + 3^2(0.2) + 4^2(0.1) + 5^2(0.1)$

$E(x^2) = 0.4 + 0.8 + 1.8 + 1.6 + 2.5 = 7.1$

$E(x^2) = 7.1$

$\sigma_x^2 = E(x^2) - (E(x))^2 = 7.1 - (2.3)^2 = 1.81$

$\sigma_x^2 = 1.81$

(c) $\sigma_x = \sqrt{\sigma_x^2} = \sqrt{1.81} = 1.345362405$

$\sigma_x = 1.345$

(d)

y	10	20	30	40	50
P(y)	0.4	0.2	0.2	0.1	0.1

(e) $\mu_y = 10(0.4) + 20(0.2) + 30(0.2) + 40(0.1) + 50(0.1)$

$\mu_y = 23$

(f) $E(y^2) = 10^2(0.4) + 20^2(0.2) + 30^2(0.2) + 40^2(0.1) + 50^2(0.1)$

(g) $E(y^2) = 710$

$\sigma_y^2 = E(y^2) - (E(y))^2 = 710 - 529 = 181$

$\boxed{\sigma_y^2 = 181}$

(g)

$\sigma_y = \sqrt{\sigma_y^2} = \sqrt{181} = 13.45$

$\boxed{\sigma_y = 13.45}$

(4)

(a) $P(x_3) = 0.1 + 0.2 + 0.4 + 0.2 + 0.1 = 1$

P_3 is a possible probability mass function of x because total of probabilities for all possible values of x equals to 1.

(b)

$\mu_x = 0(0.1) + 1(0.2) + 2(0.4) + 3(0.2) + 4(0.1)$

$\mu_x = 0.2 + 0.8 + 0.6 + 0.4 = 2.0$

$\boxed{\mu_x = 2.0}$

$\sigma_x^2 = E(x^2) - (E(x))^2$

$E(x^2) = 0^2(0.1) + 1^2(0.2) + 2^2(0.4) + 3^2(0.2) + 4^2(0.1)$

$= 0.2 + 1.6 + 1.8 + 1.6 = 5.2$

$\sigma_x^2 = 5.2 - 4 = 1.2$

$\boxed{\sigma_x^2 = 1.2}$

(5)

(a)

X	1	2	3	4	5
P(x)	0.7	0.15	0.10	0.03	0.02

(b) $P(X \leq 2)$:-

$P(X \leq 2) = P(X=1) + P(X=2) = 0.7 + 0.15$

$\boxed{P(X \leq 2) = 0.85}$

(c) $P(X > 3)$:-

$P(X > 3) = P(X=4) + P(X=5) = 0.03 + 0.02$

$\boxed{P(X > 3) = 0.05}$

(d) μ_x :-

$\mu_x = 1(0.7) + 2(0.15) + 3(0.10) + 4(0.03) + 5(0.02)$

$\mu_x = 0.7 + 0.3 + 0.3 + 0.12 + 0.1 = 1.52$

$\boxed{\mu_x = 1.52}$

(e)

$$\sigma_X = \sqrt{\sigma_X^2} = \sqrt{E(X^2) - (E(X))^2}$$

$$E(X^2) = 1^2(0.7) + 2^2(0.15) + 3^2(0.1) + 4^2(0.03) + 5^2(0.02)$$

$$E(X^2) = 0.7 + 0.6 + 0.9 + 0.48 + 0.5 = 3.18$$

$$\sigma_X^2 = 0.8696 \Rightarrow \boxed{\sigma_X = 0.9325}$$

(f)

X	24	25	26	27	28
$P(x)$	0.0825	0.0744	0.7372	0.0541	0.0518

(a) $\mu_X = 24(0.0825) + 25(0.0744) + 26(0.7372) + 27(0.0541) + 28(0.0518)$

$$\mu_X = 1.98 + 1.86 + 19.1672 + 1.4607 + 1.4504$$

$$\boxed{\mu_X = 25.9183}$$

(b) $\sigma_X = \sqrt{\sigma_X^2} = E(X^2) - (E(X))^2$

$$E(X^2) = 24^2(0.0825) + 25^2(0.0744) + 26^2(0.7372) + 27^2(0.0541) + 28^2(0.0518)$$

$$E(X^2) = 47.52 + 46.5 + 498.3472 + 39.4389 + 40.612$$

$$E(X^2) = 672.4173$$

$$\sigma_X^2 = 0.65902511 \Rightarrow \sigma_X = 0.81180361$$

$$\boxed{\sigma_X = 0.8118}$$

(6)

(7)

(a) $P(x) = \begin{cases} Cx & \text{for } x = 1, 2, 3, 4 \text{ or } 5 \\ 0 & \text{otherwise} \end{cases}$

$C \Rightarrow \quad P(x) = C(1) + 2C + 3C + 4C + 5C = 15C$

$P(x) = 1 = 15C$

$\boxed{C = 1/15}$

(b) $P(x = 2) :-$

$\boxed{P(x=2) = 2/15}$

(c) $\mu_x = 1(1/15) + 2(2/15) + 3(3/15) + 4(4/15) + 5(5/15)$

$\mu_x = 1/15 + 4/15 + 9/15 + 16/15 + (25/15)$

$\boxed{\mu_x = 3.6667}$

(d) $\sigma_x^2 = E(x^2) - (E(x))^2$

$E(x^2) = 1^2(1/15) + 2^2(2/15) + 3^2(3/15) + 4^2(4/15) + 5^2(5/15)$

$= 1/15 + 8/15 + 27/15 + 64/15 + 125/15 = \frac{225}{15}$

$\boxed{E(x^2) = 15}$

$\boxed{\sigma_x^2 = 1.5556}$

(d) $\sigma_x = \sqrt{\sigma_x^2} = \sqrt{1.5556} = 1.247219129$

$\boxed{\sigma_x = 1.2472}$

(8)

x	0	1	2	3	4
F(x)	0.41	0.72	0.85	0.95	1.00

(a) $P(X \leq 2)$:-

$\boxed{F(2) = 0.82 = P(X \leq 2)}$

(b) $P(X > 3)$:-

$P(X>3) = 1 - P(X \leq 3) = 1 - F(3)$

$P(X>3) = 1 - 0.95 = 0.05$

$\boxed{P(X>3) = 0.05}$

(c) $P(X = 1)$:-

$P(X=1) = F(1) - P(X=0)$

$P(X=1) = 0.72 - 0.41 = 0.31 \Rightarrow \boxed{P(X=1) = 0.31}$

(d) $P(X=0)$:-

$\boxed{P(X=0) = 0.41}$

(e) $P(X=0)$ is the most probable number of errors to be detected.

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(a) $f(t) = \begin{cases} 0.1\, e^{-0.1t} & t \geq 0 \\ 0 & t < 0 \end{cases}$

(a) $\mu = 1/0.1 = 10$

$\boxed{\mu = 10}$

(b) $\sigma_x^2 = 1/\lambda^2 = \frac{1}{D\text{-}P} = \frac{1}{0.01} = 100$

$\sigma_x = \sqrt{\sigma_x^2} = \sqrt{100} = 10$

$\boxed{\sigma_x = 10}$

(c) $F(t) = \begin{cases} 0 & t < 0 \\ 1 - e^{-0.1t} & t \geq 0 \end{cases}$

(d) $P(x < 12) = \int_0^{12} 0.10\, e^{-0.1t}\, dt$

$= 0.10 \left[\frac{e^{-0.1x}}{-0.1} \right] \Big|_0^{12}$

$= -1 \left[e^{-0.1(12)} - e^0 \right]$

$= 1 - 0.301194212 = 0.6988$

$\boxed{P(x < 12) = 0.6988}$

(9)

(19)

(a) $$\begin{cases} \frac{3}{64}x^2(4-x) & 0<x<4 \\ 0 & \text{otherwise} \end{cases}$$

(a) $$P(x>3) = \int_3^4 \frac{3}{64}x^2(4-x)\,dx$$

$$= \frac{3}{64}\left[\frac{4x^3}{3} - \frac{x^4}{4}\right]\Big|_3^4$$

$$= \frac{3}{64}\left[\frac{256}{3} - 64\right] - \frac{3}{64}\left[\frac{108}{3} - 20.25\right]$$

$$= \frac{3}{64}\left[\frac{64}{3}\right] - \frac{3}{64}\left[\frac{47.25}{3}\right]$$

$$= 1 - 0.73828125 = 0.26171875$$

$$\boxed{P(x>3) = 0.26172}$$

(b) $$P(2<x<3) = \int_2^3 \frac{3}{64}x^2(4-x)\,dx$$

$$= \frac{3}{64}\left[\frac{4x^3}{3} - \frac{x^4}{4}\right]\Big|_2^3$$

$$= \frac{3}{64}\left[\frac{108}{3} - \frac{81}{4}\right] - \frac{3}{64}\left[\frac{32}{3} - \frac{16}{4}\right]$$

$$= 0.73828125 - 0.3125 = 0.42578125$$

$$\boxed{P(2<x<3) = 0.42578}$$

(c) $\mu_x = \int_0^4 x \cdot \frac{3}{64} x^2(4-x)\,dx$

$$\mu_x = \frac{3}{64}\left[\frac{4x^4}{4} - \frac{x^5}{5}\right]\Big|_0^4$$

$$= \frac{3}{64}\left[x^4 - \frac{x^5}{5}\right]\Big|_0^4 = \frac{3}{64}\left[256 - 204.8\right]$$

$$= 2.4$$

$$\boxed{\mu_x = 2.4\ \%}$$

(d) $\sigma_x^2 = E(x^2) - (E(x))^2$

$$E(x^2) = \int_0^4 x^2 \cdot \frac{3}{64} x^2(4-x)\,dx$$

$$= \frac{3}{64}\left[\frac{4x^5}{5} - \frac{x^6}{6}\right]\Big|_0^4$$

$$= \frac{3}{64}\left[819.2 - 682.6667\right]$$

$$= 6.4$$

$$\sigma_x^2 = 6.4 - 5.76 = 0.64$$

$$\boxed{\sigma_x^2 = 0.64}$$

(11)

(e) $$F(x) = \begin{cases} 0 & x < 0 \\ x^3/16 - 3x^4/256 & 0 \le x < 4 \\ 1 & x \ge 4 \end{cases}$$

(20)

(a) $P(x < 0.02) =$

$$P(x < 0.02) = \int_0^{0.02} 625\, x\, dx$$

$$= 625 \frac{x^2}{2}\Big|_0^{0.02} = \frac{625}{2}(0.02)^2$$

$$P(x < 0.02) = 0.125$$

(b)

$$\mu_x = \int_0^{0.04} 625 x^2 dx + \int_{0.04}^{0.08} 50x - 625x^2\, dx$$

$$\mu_x = 625\frac{x^3}{3}\Big|_0^{0.04} + \left(\frac{50x^2}{2} - \frac{625x^3}{3}\right)\Big|_{0.04}^{0.08}$$

$$\mu_x = 0.01333 + (0.16 - 0.106667) - (0.040 - 0.013333)$$

$$\mu_x = 0.01333$$

$$\mu_x = 0.0133$$

(c) $\sigma_x = \sqrt{\sigma_{x^2}} = E(x^2) - (E(x))^2$

$$E(x^2) = \int_0^{0.04} x^2 625x\, dx + \int_{0.04}^{0.08} 50x^2 - 625x^3\, dx$$

(12)

$$= 625\left[\frac{x^4}{4}\right]\Big|_0^{0.04} + \left(\frac{50x^3}{3} - \frac{625x^4}{4}\right)\Big|_{0.04}^{0.08}$$

$$= 0.0004 + (0.0085333 - 0.0064) - (0.00106667 - 0.0004)$$

$$= 0.00106666$$

$$\sigma_x^2 = 0.00088976$$

$$\sqrt{(\sigma_x^2)} = 0.02982913$$

(23) $$f(x) = \begin{cases} \frac{3}{52}x(6-x) & 2 < x < 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) $$P(X < 2.5) = \int_2^{2.5} \frac{3}{52}x(6-x)\,dx$$

$$= \frac{3}{52}\left[\frac{6x^2}{2} - \frac{x^3}{3}\right]\Big|_2^{2.5}$$

$$= \frac{3}{52}\left[18.75 - 5.208333\right] - \frac{3}{52}\left(12 - \frac{8}{3}\right)$$

$$= 0.78125 - 0.538461538 = 0.242788462$$

$$P(X < 2.5) = 0.2428$$

(b) $P(2.5 < x < 3.5) = \int_{2.5}^{3.5} \frac{3}{52} x(6-x)\,dx$

$\boxed{P(2.5 < x < 3.5) = 0.5144}$

(c) $\mu_x = \int_2^4 x \cdot \frac{3}{52} x(6-x)\,dx$

$= \frac{3}{52} \int_2^4 6x^2 - x^3\,dx$

$= \frac{3}{52} \left[\frac{6x^3}{3} - \frac{x^4}{4} \right] \Big|_2^4$

$\boxed{\mu_x = 3}$

(d) $\sqrt{\sigma_x^2} = \sqrt{E(x^2) - (Ex)^2} = \sigma_x$

$E(x^2) = \int_2^4 x^3 \frac{3}{52}(6-x)\,dx = \frac{3}{52} \int_2^4 (6x^3 - x^4)\,dx$

$= \frac{3}{52} \left(\frac{6x^4}{4} - \frac{x^5}{5} \right) \Big|_2^4$

$\sqrt{\sigma_x^2} = \sqrt{E(x^2) - 9} = 0.5684$

$\boxed{\sigma_x = 0.5684}$

(e) $P(2.4316 < x < 3.5684) = \int_{2.4316}^{3.5684} \frac{3}{52}(6x - x^2)\,dx$

$= 0.5832$

(f)

$$F(x) = \begin{cases} 0 & x < 2 \\ (-x^3 + 9x^2 - 28)/52 & 2 \le x < 4 \\ 1 & x \ge 4 \end{cases}$$